US Opioid Overdose & Overdose Deaths An Exploration of Indicators and emergent preventative actions by state

Insights from Machine Learning Models

IST 707, August 22nd 2025

Waleed Jamil

Introduction

Opioid overdose deaths have remained a critical public health crisis in the United States since the early 2000s[1,2], with marked variation across states and demographic groups. This project develops a predictive model to estimate future overdose death rates by state and evaluates the alignment between projected risk and naloxone dispensing. Using Random Forest regression and clustering techniques, we identify jurisdictions with potential gaps in naloxone coverage. The goal is to provide actionable insights for key stakeholder agencies such as ORS and SAMHSA to optimize resource allocation and reduce preventable deaths.

While tighter prescribing regulations are already managed by the CDC[1] and NIDA[3] and enforced by the DEA[4], our findings suggest that overdose deaths will persist unless the underlying demand for illicit opioids is addressed. Illegally manufactured fentanyl (IMF) emerged as the most significant contributor to overdose mortality, with male gender and state-level factors amplifying risk—underscoring the need for targeted, multifaceted interventions beyond prescribing controls.

Literature Review

Opioid overdose deaths have surged in the United States since the early 2000s, prompting extensive public health responses from agencies such as the CDC[1], NIDA[2], and DEA[3]. These efforts include tighter prescribing regulations, expanded naloxone access, and enhanced surveillance of drug-related mortality. A recent CDC report found that six states experienced worsening overdose rates, while 25 states remained at persistently high levels[6,7]. Prior studies have emphasized demographic and geographic disparities in overdose risk, with illegally manufactured fentanyl (IMF) increasingly driving fatal outcomes[4]. Although naloxone distribution has expanded nationally, recent analyses suggest a misalignment between overdose burden and resource allocation[5]. Predictive modeling and clustering techniques have been proposed to identify high-risk jurisdictions and guide targeted interventions.

Data

All datasets (Table 1) were sourced from publicly available government repositories and contained no missing values, ensuring high reliability for modeling. As shown in Figure 1, opioids dominate the overdose landscape and interventions targeting opioids may have outsized impact. Unemployment rates were changing dramatically in the years leading up to the period under study for opioid deaths. In Figure 2, geographic targeting becomes clearer with four of the top five states even showing a stable although high trend. Increasing naloxone dispensing

appears to be correlated with the stability in these four states (Figure 3). The skyrocketing death rates in the District of Columbia with a levelling off naloxone dispensing appears concerning.

Table 1: Initial Exploration of the Three Key Input Datasets

Dataset Name	Source	Dimensions
SUDORS Overdose Deaths	CDC, NCHS	144 rows × 274 cols
Naloxone Dispensing	DEA ARCOS	255 rows × 6 cols
State Demographics	U.S. Census Bureau	51 rows × 10 cols
2020 National Census	U.S. Census Bureau	53 rows × 2 cols

Figure 1: Overall Correlation between different subtypes of overdoses

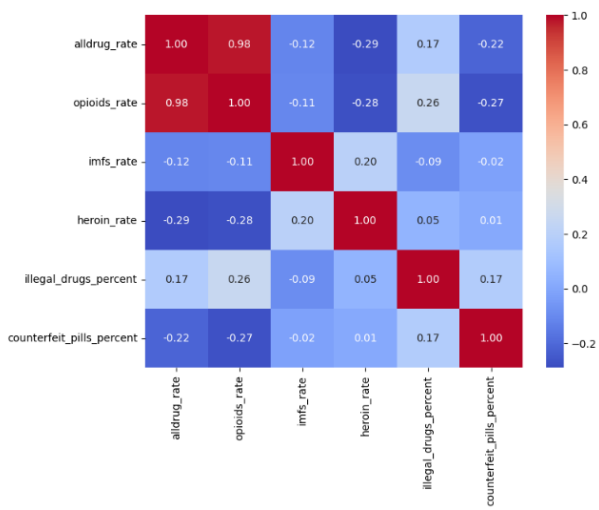


Figure 2: Top 5 States by Annual Average All Drug Death Rate

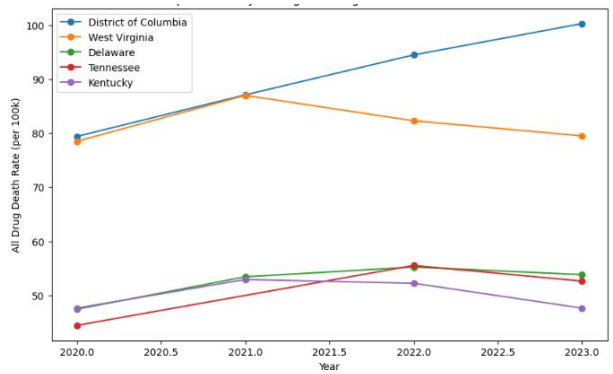


Figure 3: Differenced Monthly Unemployment Rate – Top 5 States by average All Drug Death Rate

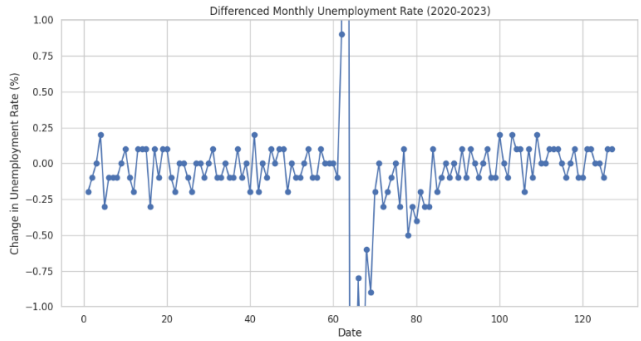


Figure 4: Evaluation of Naloxone Dispensing Rate -– Top 5 States by average All Drug Death Rate

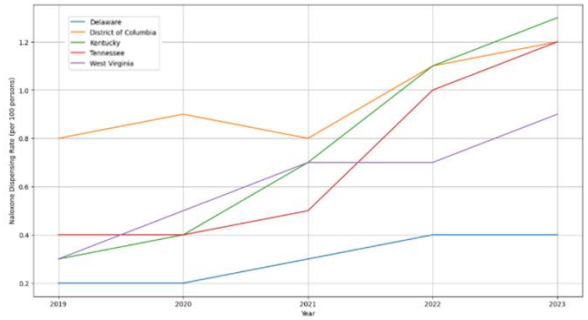


Figure 5: Top 15 Demographic Features with highest correlations with Overdose rates

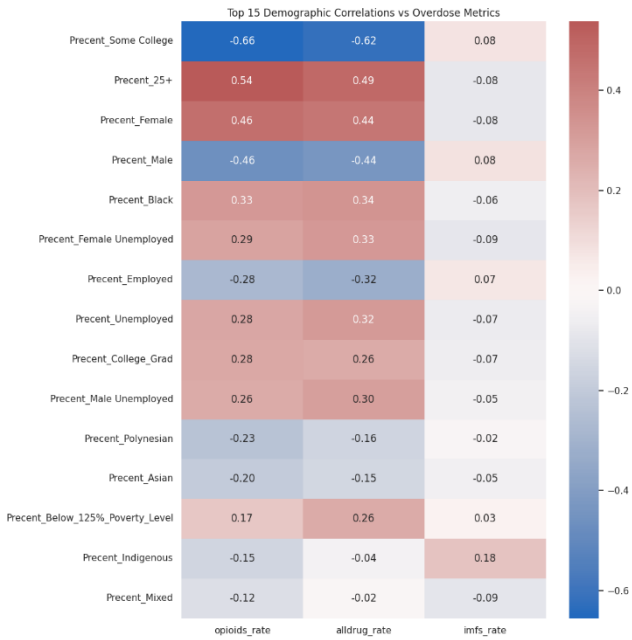
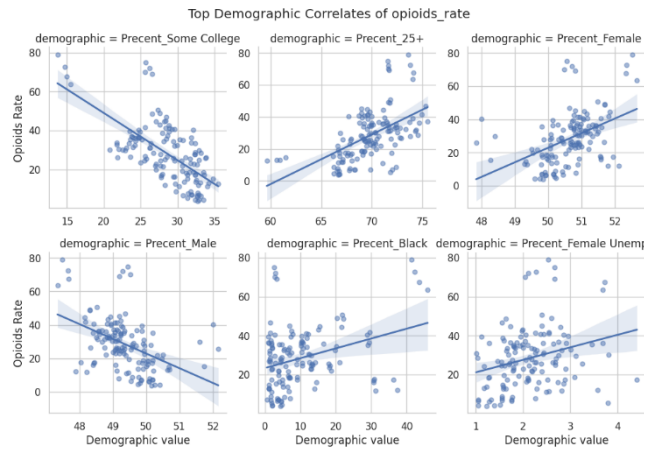


Figure 6: Plots of Top Demographic correlates with opioid rates



Methods

- **Data Sources: Integrated five datasets—**demographics, census estimates, overdose *death rates (SUDORS), naloxone dispensing rates, and unemployment statistics—across U.S. states (2020–2023).*
- ***Preprocessing:***
    - Merged datasets by state and year using left and inner joins due to missing temporal data for some states.
    - Dropped missing overdose records; imputed demographic gaps with column means.
    - One-hot encoded categorical variables; split data into train/test sets.
    - Applied KNN imputation and scaled features with StandardScaler; tested alternatives (SimpleImputer, PowerTransformer) with no performance gain.
    - Retained outliers and removed non-numeric columns for modeling.
- **Feature Engineering:**
    - Created differenced unemployment rates and naloxone dispensing metrics.
    - Aggregated drug subtype deaths; selected demographic and overdose variables as predictors.
- **Exploratory Analysis:**
    - Used descriptive statistics and correlation matrices to identify key relationships.

- Visualized overdose rates, unemployment trends, and naloxone access across states and years.

- **Modeling:**
  - Assessed opioid death correlations with demographics using multiple classifiers and regressors.
  - Selected **Random Forest** for its high accuracy ($R^2 = 0.98$); used again for naloxone and opioid death predictions ($R^2 = 0.99$ and 0.67, respectively).
  - Applied **K-means clustering** to projected 2027 naloxone access and projected opioid death categories (high, medium, low).
  - Applied Linear, Random Forest, and Gradient Boosting regressors to demographic data to predict opioid deaths

Fig 7: Model evaluation for rate projections

'Comparison of Models for Opioid Overdose Rate:'

	Model	MSE	RMSE	R-squared
0	Linear Regression	116.945259	10.814123	0.502967
1	Random Forest Regressor	3.419315	1.849139	0.985467
2	Gradient Boosting Regressor	1.811605	1.345959	0.992300

'Comparison of Models for Naloxone Dispensing Rate:'

	Model	MSE	RMSE	R-squared
0	Linear Regression	6.458420	2.541342	-52.349500
1	Random Forest Regressor	0.039582	0.198951	0.673038
2	Gradient Boosting Regressor	0.042630	0.206470	0.647859

Fig 8: K-means Clustering of projected rates

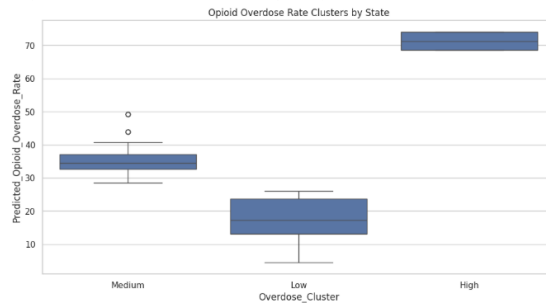


Fig 9: Model evaluations for demographic data models

Demographic-only models predicting: opioids_rate

	Model	CV_R2_Mean	CV_R2_SD	Test_R2	Test_RMSE
0	GradientBoosting	0.041539	0.470130	0.394501	9.340556
1	RandomForest	0.193492	0.370682	0.336870	9.774966
2	Linear	-0.073322	0.236679	0.015755	11.908792

Fig 10: Examples of missing temporal data for naloxone dispensing by state

'Years each state is missing in SUDORS data:'

Year	2020	2021	2022	2023
Missing State				
Alabama	2020	2021	2022	2023
Arkansas	2020	<NA>	<NA>	2023
California	2020	2021	2022	2023
Florida	2020	2021	2022	2023
Hawaii	2020	<NA>	<NA>	<NA>
Idaho	2020	2021	2022	2023

- **Validation:**
  - Evaluated models using MSE, R^2 scores, and 5-fold cross-validation.
  - Used Mean Decrease Impurity (MDI) to identify IMF-related deaths as the strongest predictor.

Results

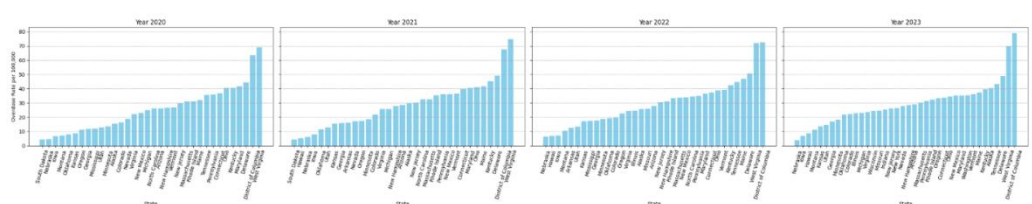
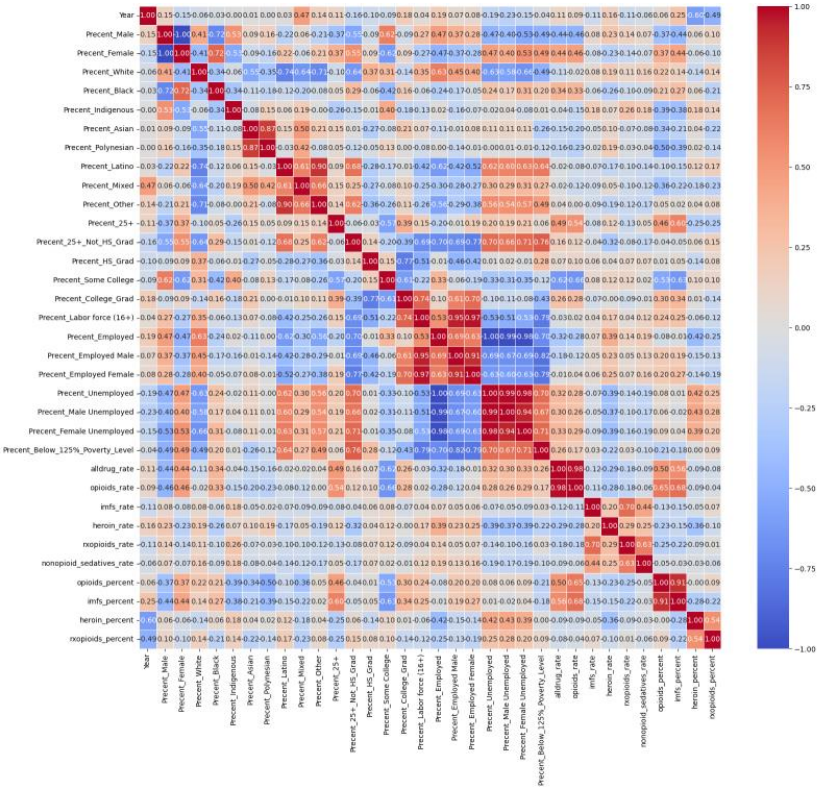


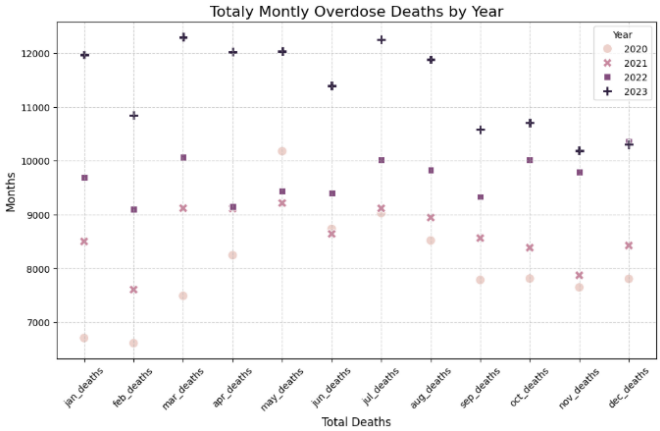
Figure 11: Geographic disparities – identification of consistently high-burden states

Figure 12: Correlation Matrix of Overdose Subtypes & Demographics



- Strong correlation between **opioid-related deaths** and **total drug overdose deaths**, suggesting opioids are the primary driver of national overdose trends.
- There were also relatively strong correlations between the percentage of a population without a high school diploma or equivalent and unemployment, as well as unemployment and percent of population below 125% of the poverty level. Neither of these were correlated with overdose death rates, however.

Figure 13: Total Monthly Overdose Deaths by Year



- Although we did not have a large set of pre-COVID data, there is a noticeable spike in the total number of overdoses in the immediate aftermath of the lockdown (May 2020). The total number of overdose deaths then decreased slightly and have since continued to increase to the current elevated level.

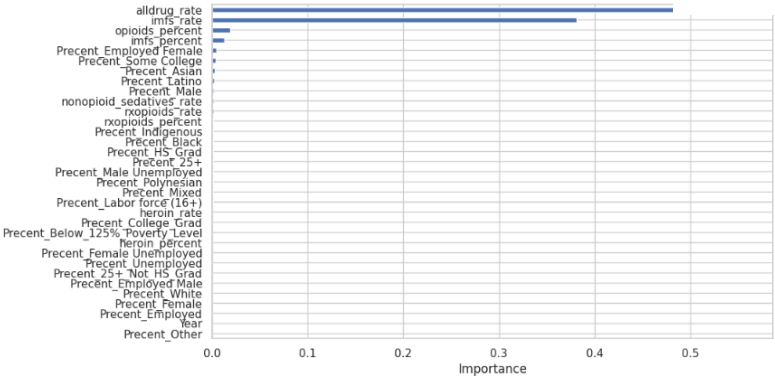


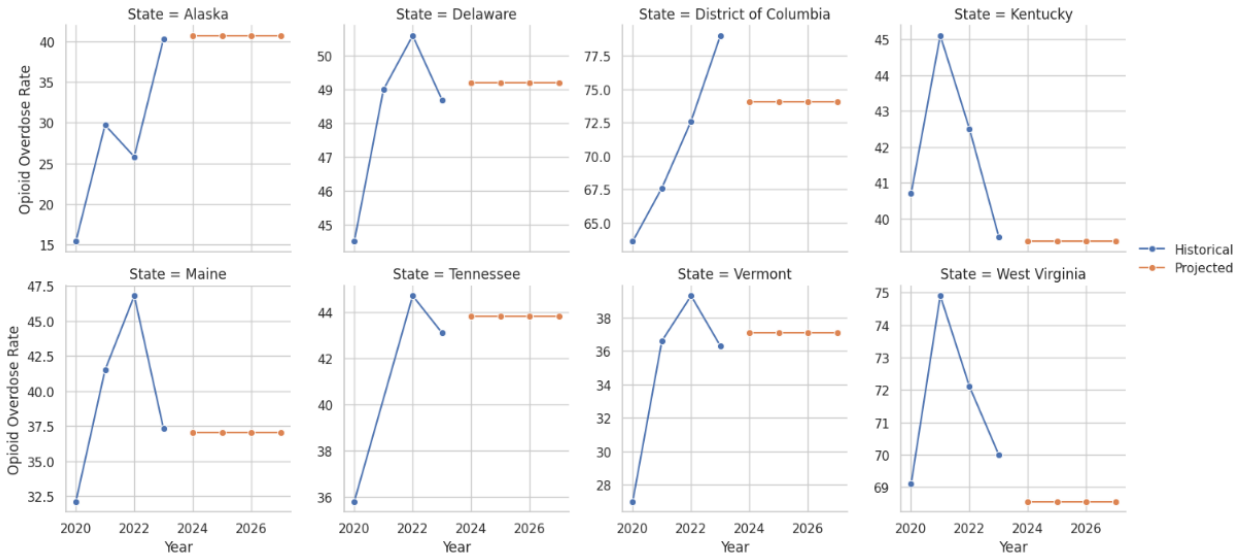
Figure 14: Feature Importance of Overdose Subtypes & Demographics in determining Opioid Deaths

- Demographic factors like poverty level, education, and unemployment showed weaker influence on the model's predictive

ability than expected, challenging initial assumptions.

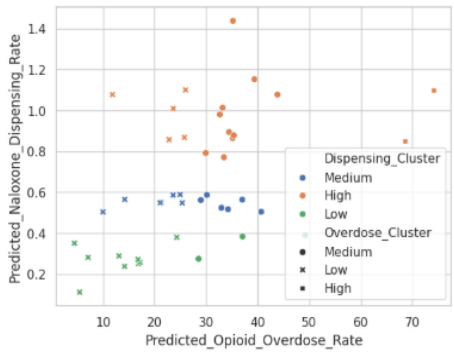
- **IMFs rate** (illegally manufactured fentanyl) is the **most influential predictor** of opioid mortality.
- States like West Virginia, DC, Kentucky, Tennessee, and Delaware consistently ranked highest in overdose mortality as shown in Figure 2.
- Monthly overdose deaths showed **seasonal variation**, with peaks in summer and early fall.

Figure 15: Projected Opioid Overdose rates indicate states like Alaska with increased need for targeted intervention



- Predicted naloxone dispensing rates for 2027 showed **moderate accuracy** ($R^2 = 0.67$), useful for identifying future mismatches.
- K-means clustering categorized states into **high, medium, and low risk** for projected 2027 opioid deaths and naloxone access.
- Naloxone dispensing rates were **misaligned** with overdose burden in several high-risk states, revealing gaps in intervention coverage.
- Continuous Risk Index used projections to calculate risk of worsening opioid crisis; **West Virginia, Delaware, and Washington DC** were associated with the highest risk.

Figure 16: Clustered levels of Opioid Deaths by State mismatched with Naloxone Dispensing levels (Projected 2027)



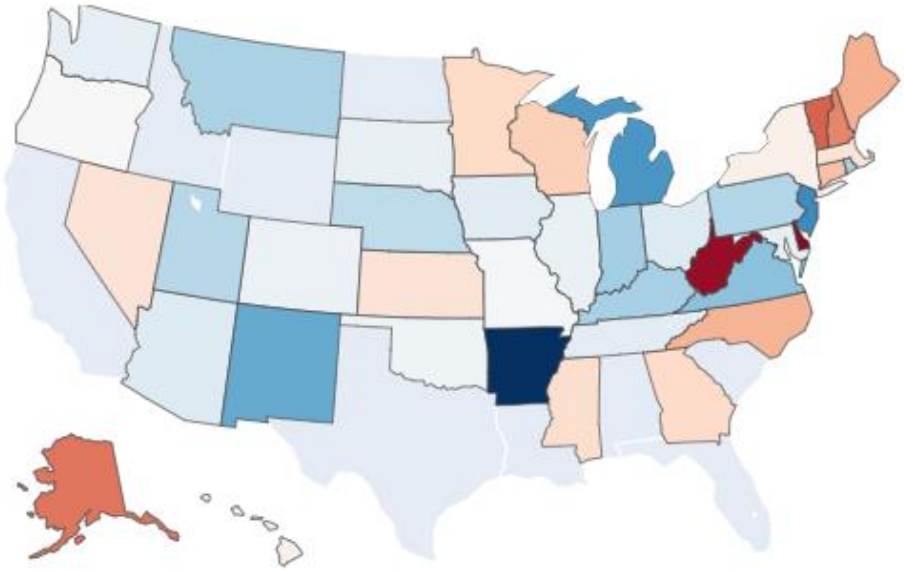


Figure 17: Continuous Risk Index (Projected 2027) – Overdose Deaths vs. Naloxone Dispensing (Red: High Risk, Blue: Low Risk)

Discussion

This analysis confirms that opioid-related deaths—particularly those involving illegally manufactured fentanyl (IMF)—are the primary driver of overall drug overdose mortality in the U.S. The Random Forest model achieved high predictive accuracy ($R^2 = 0.98$), with IMF-related deaths emerging as the most influential feature. Contrary to expectations, socioeconomic indicators such as poverty, education, and unemployment showed weaker correlations with overdose rates, suggesting that structural factors alone may not fully explain state-level disparities. It was also interesting to note that per the correlation matrix, the rate of drug overdose deaths involving illegally made fentanyl was surprisingly more strongly correlated with the rate of overdose deaths involving prescription opioids ($R^2 =0.69$) than the rate involving heroin (R^2 0.38). This suggests that prescription drug abuse could be driving the increase in fentanyl use and ultimately overdose deaths.

Naloxone dispensing rates were not consistently aligned with overdose burden, revealing critical gaps in intervention coverage. Geographic clustering and forecasting for 2027 further highlighted states at elevated risk for opioid deaths and under-dispensing of naloxone. These mismatches underscore the need for targeted resource allocation and policy adjustments. However, no matter how much naloxone is distributed, there will always be overdose deaths until the demand for illicit drugs is decreased.

Seasonal trends in overdose deaths suggest temporal spikes that may inform timing of public health campaigns. Overall, the findings support prioritizing fentanyl-focused interventions and refining naloxone distribution strategies to better match projected needs. The demographics data's prediction for opioid death rates was weak, indicating that demographics themselves are not a great indicator of opioid death rate. Based on the results, factors like rurality, access to treatment, and naloxone availability may play a key role and may be more impactful than demographics alone.
The continuous risk index, which was calculated using the z-score mismatches between projected naloxone dispensing and overdose deaths, indicated that **states including West Virginia and Delaware, along with the city of Washington DC are in the greatest need for resource or policy realignment, since the overdose death are projected to rise with naloxone dispensing rates projected to be insufficient.**

Limitations

1. **Input Data limitations**: The projections and analysis are based on limited temporal datapoints and may not capture long term trends or baseline pre-COVID. State-level aggregation may mask local disparities; neighborhood or county-level data could reveal more targeted insights. Naloxone dispensing rates may not reflect actual usage or access at the time of overdosing, limiting interpretation of its effectiveness. Data for several states were also completely unavailable, and our analysis included 135 rows, out of a total of a possible 208 rows (4 years * 52 states, DC and PR included).
2. **Imputation and modeling**: Imputed demographic values may introduce bias, especially in states with sparse reporting. While Random Forest performed well at the state level, results may not generalize in future years and to local trends.
3. **Missing predictors**: Factors like mental health access and illicit drug supply metrics were not included but may influence outcomes.

Future Work

Future work will focus on expanding both the temporal scope and feature set by incorporating additional years, more granular predictors, and localized data sources. Integrating real-time signals—such as emergency call records, prescription monitoring data, and social media indicators—will enhance the robustness and responsiveness of predictive models.

A key objective is to embed geographic and temporal dynamics into forecasting frameworks to better identify overdose hotspots and seasonal risk patterns. Additionally, the scope of intervention modeling will be broadened beyond naloxone dispensing to include buprenorphine access, public information campaigns, and other evidence-based strategies aimed at reducing opioid-related harm. Analyzing data from each state on the prescription rate of opioids through the years in order to determine it possible impact on overdose deaths would also be of interest.

References

---

[1] CDC focuses on preventing harms related to drug use and overdose through various strategies.

[2] The National Center for Health Statistics (NCHS) is the United States' principal health statistics agency. It is part of the Centers for Disease Control and Prevention (CDC). NCHS collects, analyzes, and disseminates timely, relevant, and accurate health data and statistics to guide health policies and programs.

[3] The National Institute on Drug Abuse (NIDA) is a federal agency leading research on drug abuse for prevention, treatment, and public health.

[4] The Drug Enforcement Administration (DEA) enforces drug laws and works to disrupt the illicit drug supply.

[5] The National Academy of Medicine (NAM) has an initiative to develop solutions for reducing substance misuse.

[6] https://www.cdc.gov/overdose-prevention/about/index.html#:~:text=However%2C%20drug%20overdose%20deaths%20declined,made%20fentanyl%20and%20fentanyl%20analogs.

[7] https://www.cdc.gov/nchs/products/databriefs/db522.htm

[8] https://www.cdc.gov/overdose-prevention/data-research/facts-stats/sudors-dashboard-fatal-overdose-data.html?utm_source=chatgpt.com

[9] https://www.cdc.gov/overdose-prevention/data-research/facts-stats/us-dispensing-rate-maps.html

[10] https://www.cdc.gov/overdose-prevention/reversing-overdose/about-naloxone.html

[11] https://www2.census.gov/programs-surveys/popest/datasets/2020-2024/state/totals/

[12] https://data.bls.gov/timeseries/LNS14000000